

REC: Accelerating Sustainable Energy Sector Growth with the Power of AI

Artificial Intelligence and Machine Learning- the imitation by computers and robots of the intelligence inherent in humans for planning, decision making, improved communication and processing large amounts of data- is significantly changing industries across the world.

Standing at the forefront of India's dynamic power sector, REC is proactively integrating these cutting-edge technologies to enhance internal operations while also leading the integration of AI/ML across India's power distribution sector.



Jitendra Srivastava
Chairman & Managing Director
REC Ltd.

Vision- Driven Investments

REC's strategic investment and growth is driven by its deep commitment to aligning its funding with the nation's broader development goals. Driven by its Maharatna status, REC is aggressively targeting a massive expansion of its total Loan Book from the current ₹5.82 lakh crore to a landmark ₹10 lakh crore by FY2030.

REC is also prioritizing Renewable Energy (RE) financing and aims to be a leader in India's energy transition, targeting a loan outstanding of ₹3 lakh crore (30% of its business) by 2030.

The pace of this comprehensive expansion is further accelerated by the deep integration of cutting-edge technology across all operations.

Data-Informed ESG Excellence

REC has cemented its commitment to sustainability by achieving the top-rank in the latest NSE ESG Ratings. This outstanding performance is a direct result of a comprehensive three-year strategy initiated with the company's first-ever ESG Policy, guided by the REC Board of Directors.

Key drivers of this success include strategic planning, benchmarking against global standards like Climate Disclosure Project (CDP), S&P CSA, and developing a clear Net-Zero Pathway.

Looking ahead, REC is further ensuring its sustained ESG leadership by leveraging data analytics and AI to drive smarter, more sustainable decision-making across operations.

Digital Transformation with RDSS

A core component of the power sector's digital transformation is the deployment of smart meters under the Revamped Distribution Sector Scheme (RDSS). REC and PFC are the nodal agencies for facilitating the implementation of the scheme. More than 40 million smart meters have been deployed across India.

A Future-Ready Energy Sector

Further advancing the application of data-driven intelligence tools within the power sector, REC, under the aegis of the Ministry of Power and in association with PFC Limited., conducted a two-day National Conference on the Use of AI/ML in the Power Distribution Sector under the Revamped Distribution Sector Scheme (RDSS) on December 6-7, 2025, at Bharat Mandapam, New Delhi.

The Conference showcased cutting-edge use cases of smart meter data analytics, integrated IT/OT systems, and Smart Home Automation Solutions that drive innovation, improve operational efficiencies, and enhance consumer satisfaction across Distribution Companies (DISCOMs) in the country.

Shri Manohar Lal, Hon'ble Union Minister of Power and Housing & Urban Affairs, graced the event and noted that the that the AI/ML based solutions, smart meter analytics, digital twins, predictive maintenance, theft detection intelligence, appliance-level consumer insights, automated outage prediction and GenAI-based decision support can transform both consumer experience and operational efficiency.

Strengthening Digital Infrastructure with India Energy Stack (IES)

Another major stride towards leveraging AI to transform the power sector has been the initiation of the India Energy Stack (IES) by the Ministry of Power. REC is the Nodal Agency for IES.

The India Energy Stack, defined as a DPI, aims to identify and connect stakeholders and assets, and facilitate open data exchange through uniform specifications and standards, thereby unlocking transparent, reliable, inclusive, efficient, and affordable energy access. The project is scheduled to be completed by July 2026.

Driving the Energy Evolution

From powering India's Green Revolution by energising agricultural pump sets to championing the rise of green technologies in the energy ecosystem, REC has remained driven by one consistent purpose- meeting the nation's needs today while preparing it for the demands of tomorrow.

As the company continues to play a pivotal role in advancing the power sector, REC is actively exploring emerging innovations in Artificial Intelligence and harness their potential to shape a smarter, more resilient and future-ready energy landscape.